Natural Language Processing

The Programming Historian – creating word embeddings

2025/26

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1. For this week's lab session we will follow a lesson and python notebook from the https://programminghistorian.org/
   (a) Open the lesson here in your web browser: https://programminghistorian.org/en/lessons/understanding-creating-word-embeddings
   (b) To use the Jupyter notebook provided, make sure you have Jupyter installed

       pip install jupyter

   (c) To use the Jupyter notebook in VSCode, also install the VSCode Jupyter extension. (https://code.visualstudio.com/docs/datascience/jupyter-notebooks)
2. The lesson walks through the process of creating word embeddings from a corpus using the gensim library
   (a) Install the gensim library if it is not already present

       pip install --upgrade gensim

   (b) There is a bug that causes gensim installation to fail with some versions of SciPy. If you run into this problem on your own machine you can install gensim by going back to the previous version of SciPy:

       pip install scipy==1.12

   (c) Guides to working with Python on Windows and Mac machines https://docs.python.org/3/using/windows.html, https://docs.python.org/3/using/mac.html
   (d) Documentation for pip https://pip.pypa.io/en/stable/getting-started/
   (e) A guide to the built-in virtual environment module venv is here: https://realpython.com/python-virtual-environments-a-primer/#how-can-you-work-with-a-pytho
   (f) There is a new popular package manager for Python called **uv**. Try it if you are having problems (e.g. with conda): https://docs.astral.sh/uv/guides/install-python/.
3. Download the corpus linked in the lesson, extract it, and note its location on your computer

(a) The link to download the corpus is at the beginning of https://programminghistorian.org/en/le
creating-word-embeddings section of the lesson.

(b) If this link is not workign, the corpus is also available on the course Moodle page.

4. Work through the full lesson. You can do this by running the cells of the Jupyter notebook, or by using the standalone python file provided on the Moodle.